APPLICATION SUMMARY · V1.0

Regulation Station

Nervous System Operations Dashboard for WFH Solopreneurs

STACK	ARCHITECTURE	BACKEND
React 19 + Vite 7 + Tailwind 3	SPA · PWA · Offline-first	Supabase (optional)

THEORY	GENERATED
Polyvagal Theory	2026-03-05

- Frozen/Shutdown
- Anxious/High-Alert
- Safe/Flow

01 Overview

Regulation Station is a calm-tech single-page application built on Polyvagal Theory. It helps work-from-home solopreneurs identify their current nervous system state (Frozen, Anxious, or Flow), then apply targeted somatic protocols, curated task lists, and binaural audio to shift toward regulated, productive work.

The app is fully offline-capable using `localStorage`. An optional Supabase backend enables cross-device session syncing, authentication, and team collaboration features.

COMPONENTS	CUSTOM HOOKS	THEMES
32	10+	3
Specialized UI modules	Logic abstracted from UI	Dark · Light · Pastel

STATES	BUILD SIZE	DEV PORT
3	~456kB	5173
Frozen · Anxious · Flow	JS + 25kB CSS (gzipped)	`npm run dev`

02 Nervous System States

All state logic is centralized in `src/data/stateData.js` — a single `STATES` object keyed by `'frozen' | 'anxious' | 'flow'`. Each state defines its accent color, reset protocol, task list, binaural audio configuration, and tips.

Frozen / Shutdown

`#c4604a`

Dissociated · Low energy · Can't start

Protocol: Ear-Apex Pull (60 sec, 5 steps)

Audio: Theta Emergence — 5 Hz (180 + 5 Hz)

Tasks: Inbox, water, file, walk, list

Dorsal vagal shutdown — parasympathetic brake fully engaged, immobility response.

Anxious / High-Alert

`#c8a040`

Scattered · Overthinking · Chest tight

Protocol: Rib-Cage Expansion (60 sec, 5 steps)

Audio: Alpha Anchor — 10 Hz (200 + 10 Hz)

Tasks: Draft, Loom, outline, respond, review

Sympathetic mobilization — fight/flight response, elevated arousal, tunnel vision.

Safe / Flow

`#52b87e`

Grounded · Present · Ready to build

Protocol: Peripheral Vision Soften (60 sec, 4 steps)

Audio: Gamma Forge — 40 Hz (200 + 40 Hz)

Tasks: Create, strategy, calls, build, learn

Ventral vagal engagement — social safety system, safe and connected, peak cognition.

03 Architecture

App Flow

1. **Init** — ThemeProvider wraps app; check `vagaFirstVisitComplete` in localStorage; load sessions; calculate streak.
2. **First Visit** — Show `FirstVisitExperience` (30-sec breathing timer + state intro). On complete, set localStorage flag.
3. **Main Layout** — Header (fixed) → StateAssist → StateSelector → DailySummary → HRV Indicator → (if state selected) Dashboard Grid → TeamPanel → WeeklyIntelligenceCard → Footer.
4. **Overlays** — Fixed-position layers rendered at bottom of App.jsx at various z-index levels: NeuralBackground (z-0), ImmersionBackground (z-1), content (z-10), modals (z-50 to z-100).
5. **State Transitions** — Selecting a state resets timer; switching states clears all transient overlays and task load; completing any reset triggers PostResetCheckin.

State Management

No external state library. All central state lives in `App.jsx` and is passed down as props:

`selectedState`	'frozen' | 'anxious' | 'flow' | null — drives entire UI
`panicOpen`	boolean — shows PanicReset overlay
`focusOpen`	boolean — shows FocusMode overlay
`flowLockOpen`	boolean — shows FlowLock overlay
`ruptureOpen`	boolean — shows RuptureModal
`isImmersive`	boolean — toggles immersion mode
`checkinPending`	{ source, accentHex, state } | null — triggers PostResetCheckin

Data Flow

```
stateData.js (STATES)
  └► App.jsx (selectedState → stateData = STATES[selectedState])
        ├► StateSelector (read + set selectedState)
        ├► StealthReset (stateData.reset)
        ├► TaskFilter (stateData.tasks)
        ├► AudioPlayer + useAudioEngine (stateData.audio)
        └► All overlays (accentHex, selectedState)
```

04 Components

Core Components

`App.jsx` src/App.jsx	Root orchestrator. Holds all central state, keyboard shortcuts, session loading, streak calculation, and renders the full layout including all overlays.

`Header.jsx` src/components/	Sticky header with logo, HRV gauge, immersion toggle, streak/stats display, auth section, rupture button, and real-time clock (HH:MM:SS).
`StateSelector.jsx` src/components/	3-column card grid for state selection. Shows accent colors, descriptions, Polyvagal note, and active state indicator (pulsing dot).
`StealthReset.jsx` src/components/	Primary 60-second somatic protocol timer. Shows countdown, step cues, 5-phase progress bars, activation rating (1–10), and completion tone. Includes video background and VideoModal.
`TaskFilter.jsx` src/components/	Capacity Calibrator — displays state-specific task list with cognitive load indicator. User drops tasks until 1 remains (Singular Focus). OVERLOAD → OPTIMAL transition.
`AudioPlayer.jsx` src/components/	Binaural beats player. Shows frequency info (L/R ear Hz), Lissajous canvas visualizer, progress bar, ±15s controls, volume slider, and track queue.

Protocol & Reset Components

`PanicReset.jsx` src/components/	Full-screen emergency 30-second reset with expanding breathing orb SVG. Activation capture before, completion tone, auto-proceeds to PostResetCheckin.
`FlowLock.jsx` src/components/	90-minute deep-work blocker overlay (Cmd+F in Flow state). Minimal UI, logs flowMinutes on exit. Plays completion tone at 90 min.
`FocusMode.jsx` src/components/	60-second focused work mode with timer. Linked to App.jsx state via F key shortcut.
`RuptureModal.jsx` src/components/	Rupture protocol overlay for extreme panic situations. Opened via R key or ⚠ header button.

Feedback Components

`PostResetCheckin.jsx` src/components/	Bottom-right slide-up modal after every reset. Step 1: Shift? (Worse/Same/Better/Much Better). Step 2: Activation now? (1–10). Auto-dismisses after 12 seconds.
`DailySummary.jsx` src/components/	Compact stats strip showing today's reset count, avg shift arrow direction, and flow minutes. Only renders if resetCount > 0.

`StateAssist.jsx` src/components/	2-question Energy/Focus quiz that recommends a state when none is selected. Dismissable.

Analytics & Insight Components

`HRVIndicator.jsx` src/components/	HRV proxy gauge visual. State-dependent values: 31% frozen, 42% anxious, 78% flow.
`WeeklyConsistency.jsx` src/components/	7-day session frequency visualization bar chart.
`WeeklyIntelligenceCard.jsx` src/components/	Weekly summary card showing dominant state, avg recovery time, most effective protocol, and volatility score from useSessionMetrics.
`StreakMilestone.jsx` src/components/	Celebratory notification that appears when the user hits a streak milestone.

Immersion Mode Components

`ImmersionContainer.jsx` src/components/	3-phase regulated deep-work overlay (z-100). Wraps entire immersion experience.
`NeuralBackground.jsx` src/components/	Animated procedural neural background (z-0) rendered behind all content in immersion mode.
`ImmersionBackground.jsx` src/components/	State-adaptive breath-synced pulse layer (z-1). Color driven by selected state's accentHex.
`BreathingOrb.jsx` src/components/	Animated breathing orb centered above cards in immersion mode when a state is selected.
`VagusLogSidebar.jsx` src/components/	Floating sidebar visible in immersion mode. Shows streak + session history in a muted style.
`AmbientSoundscape.jsx` src/components/	Ambient audio element that plays in immersion mode as environmental backing.

Team & Auth Components

`TeamPanel.jsx` src/components/	Create/join/leave teams with 6-char invite codes. Shows roster with member streaks and latest states (self-only session detail via RLS).

`AuthModal.jsx` src/components/	Email/password sign in and sign up modal. Calls useAuth() methods.

05 Custom Hooks

All hooks live in `src/hooks/`. They handle side effects, Web Audio, session persistence, and analytics — keeping components purely presentational.

`useSessionLog()`

Persists sessions to `localStorage` key `vaga-sessions` (max 90 entries). Syncs to Supabase `nervous_system_sessions` table when authenticated. Returns `{ sessions, logSession, clearSessions }`. Session shape includes `state, type, durationSec, resetCompleted, outcome, shift, flowMinutes, activationBefore, activationAfter`.

`useAudioEngine({ carrierHz, beatHz })`

Web Audio API binaural beat engine. Left ear = carrierHz, right ear = carrierHz + beatHz. Also generates pink noise (Paul Kellett algorithm, low-pass at 4kHz). Handles browser autoplay policy. Returns `{ playing, play, pause, volume, setVolume, supported }`.

`useStreak(sessions)`

Memoized streak calculation from session array. Extracts unique calendar days, sorts descending, counts consecutive days from most recent. Returns `{ streak, lastCheckin }`.

`useTeam()`

Team CRUD via Supabase RLS. `createTeam(name)`, `joinTeam(inviteCode)`, `leaveTeam()`, `fetchMembers(teamId)`. Returns `{ team, members, createTeam, joinTeam, leaveTeam }`. Graceful no-op if Supabase is null.

`useSessionMetrics(sessions) — pure exports`

`computeMetrics()` returns averageActivation, averageDelta, stateFrequency. `generateWeeklySummary()` returns dominantState, avgRecoveryTime, mostEffectiveProtocol, volatilityScore (proportion of state transitions across consecutive sessions).

`useBreathTimer(running)`

Cyclic breathing phase manager for animations. 13-second cycle: 4s inhale, 1s hold, 8s exhale. Uses requestAnimationFrame. Returns `{ phase, cycleProgress, cycleTimeMs }`.

`useCompletionTone()`

Plays a short Web Audio API completion beep when protocols finish.

`useHRV()`

HRV proxy calculation for the gauge indicator in the Header.

06 Context Providers

ThemeContext.jsx
src/context/

Exposes `useTheme() → { theme, setTheme }`. Persists to `localStorage: nso-theme`. Sets `data-theme` on `<html>`. CSS custom properties defined per theme in `index.css`.

AuthContext.jsx
src/contexts/

Exposes `useAuth() → { user, session, loading, signIn, signUp, signOut }`. Listens to Supabase `onAuthStateChange`. All methods return graceful errors when Supabase is null.

07 Key Features

⚡ Panic Reset

30-second emergency regulation triggered by the bottom-left ⚡ button or `Cmd+Shift+R`. Expanding/contracting SVG breathing orb synced to inhale/exhale cues. Auto-proceeds to PostResetCheckin on completion.

🔒 Flow Lock

90-minute deep-work blocker overlay for the Flow state (`Cmd+F`). Minimal distraction UI, tracks elapsed minutes, plays completion tone, logs `flowMinutes` to session log on exit.

🌿 Immersion Mode

Toggled via `I` key. Shows procedural neural background, breathing orb animation, vagus log sidebar, ambient soundscape. Hides all operational noise for distraction-free deep work.

🎧 Binaural Audio

Real Web Audio API binaural beats generated on-device. Frozen: Theta 5 Hz (180+5 Hz). Anxious: Alpha 10 Hz (200+10 Hz). Flow: Gamma 40 Hz (200+40 Hz). Pink noise backing. Lissajous canvas visualizer.

📊 Session Analytics

Daily Summary strip, 7-day consistency graph, Weekly Intelligence card with dominant state, average recovery time, most effective protocol, and volatility score. Streak tracking with milestone celebrations.

👫 Team Features

Create/join teams with 6-char invite codes. View roster with members' streaks and latest states. Own session data only visible to self (Supabase RLS). Graceful no-op when offline.

📱 PWA / Offline-First

Fully functional without Supabase using localStorage. PWA-enabled via `vite-plugin-pwa` with auto-update and Google Fonts cache-first. Installable as standalone app on desktop/mobile.

🎨 3-Theme System

Dark (default), Light, and Pastel themes via CSS custom properties on `data-theme`. Biophilic, non-neon accents: sage green, amber gold, terracotta. Persisted in localStorage.

08 Keyboard Shortcuts

KEY	ACTION
`1` / `2` / `3`	Select Frozen / Anxious / Flow state
`Cmd+1` / `Cmd+2` / `Cmd+3`	Same with modifier key
`R`	Open Rupture modal
`F`	Toggle Focus mode (requires state selected)
`I`	Toggle Immersion mode
`Cmd+F`	Toggle Flow Lock (Flow state only)
`Cmd+Shift+R`	Trigger Panic Reset
`Escape`	Close all overlays

09 Data & Persistence

localStorage Keys

KEY	PURPOSE	SHAPE
`nso-theme`	Active theme	`'dark' | 'light' | 'pastel'`
`vaga-sessions`	Session log	JSON array, max 90 entries
`vagaFirstVisitComplete`	Skip onboarding	`'true'` or absent

Session Shape

```
{
  id, date, state, type, durationSec, resetCompleted, outcome,
  shift, flowMinutes, activationBefore, activationAfter,
  activationDelta, protocolUsed, protocolStartedAt,
  protocolCompletedAt, recoveryDurationSeconds, timestamp
}
```

Supabase Tables (Optional)

TABLE	COLUMNS	NOTES
`nervous_system_sessions`	Full session shape above	RLS — user sees own rows only
`teams`	id, name, invite_code, owner_id, created_at	6-char random invite code
`team_members`	id, team_id, user_id, joined_at	Join table; owner auto-added on create

10 Tech Stack & Config

Dependencies

PACKAGE	VERSION	PURPOSE
`react`	^19.2.0	UI framework
`react-dom`	^19.2.0	DOM rendering
`vite`	^7.3.1	Build tool + HMR dev server
`tailwindcss`	^3.4.19	Utility-first CSS
`@supabase/supabase-js`	^2.98.0	Auth + realtime DB (optional)
`framer-motion`	^12.34.5	Animations
`lucide-react`	^0.576.0	Icon library
`vite-plugin-pwa`	^1.2.0	PWA manifest + service worker

Tailwind Custom Tokens

TOKEN	VALUE	USAGE
`egreen`	#52b87e	Flow state accent
`eamber`	#c8a040	Anxious state accent
`ered`	#c4604a	Frozen state accent
`charcoal-900`	#111318	Deep background panels
`shadow-green-glow`	0 0 20px rgba(82,184,126,0.3)	Flow state card glow
font-mono	JetBrains Mono	Data, labels, code
font-sans	Inter	Body, headings

PWA Manifest

`name`	VAGA OPS — Regulation Station
`short_name`	VAGA OPS
`display`	standalone
`theme_color`	#060d1a
`registerType`	autoUpdate
`Fonts caching`	CacheFirst strategy for Google Fonts

npm Commands

```
cd regulation-station/
npm run dev      # dev server → localhost:5173
npm run build    # production build → dist/  (~456kB JS + 25kB CSS)
npm run lint     # ESLint
npm run preview  # preview production build
```

Environment Variables (.env.local)

```
VITE_SUPABASE_URL=...
VITE_SUPABASE_ANON_KEY=...
```

Without these variables, `src/lib/supabase.js` exports `null` and the app runs fully offline.

11 Design System

CSS Theme Variables (Dark Default)

VARIABLE	DARK	LIGHT	PASTEL
`--bg-base`	#0f1410	#f5ede4	#e8ede9
`--bg-panel`	#161d15	#ede4db	#d8e5dc
`--border`	#263024	(lighter variants)	
`--text-primary`	#eaf0eb	(dark variants)	
`--accent-flow`	#52b87e	same across all themes	
`--accent-anxious`	#c8a040	same across all themes	
`--accent-frozen`	#c4604a	same across all themes	

Design Principles

- **Biophilic, not neon** — warm earth tones, no harsh bright colors
- **Breathe animations** — 4-second cycles that mirror physiology
- **Monospace for data** — JetBrains Mono at small sizes for precision
- **Soft glows** — state-colored radial shadows, no sharp UI borders
- **Dark default** — optimized for WFH comfort (#0f1410 background)
- **Large touch targets** — buttons minimum 2.5rem for accessibility